Started on	Wednesday, 30 September 2026, 10:21 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 10:32 PM
Time taken	10 mins 53 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include<stdio.h>
#include<stdlib.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high){
    int pivot = arr[high];
    int i = (low-1);
    for(int j=low;j<high;j++){
        if(arr[j]<pivot){
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i+1], &arr[high]);
    return (i+1);
}
void quickSort(int arr[], int low, int high){
    if(low<high){
        int pi=partition(arr,low,high);
        quickSort(arr,low,pi-1);
```

```
        quickSort(arr,pi+1,high);
    }
}
int main(){
    int n;
    if(scanf("%d",&n)!=1) return 0;
    int *arr = (int *)malloc(n*sizeof(int));
    for(int i =0;i<n;i++){
        if(scanf("%d",&arr[i])!=1) return 0;
    }
    quickSort(arr,0,n-1);
    for(int i = 0;i<n;i++){
        printf("%d",arr[i]);
        if(i<n-1){
            printf(" ");
        }
    }
    printf("\n");
    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.